

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2104

Course Title: Data Structures Lab

Assignment No: 07

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Lab Section: B2

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CSE2104 Offline-7 Report

Problem Statement / Tasks:

Implement **BFS** and **DFS** in two separate .cpp files.

Source Code (Task 1: Breadth-First Search (BFS)):

```
#include <bits/stdc++.h>
using namespace std;

void BFS(vector<int> graph[], int start, int s)
{
    vector<bool> marked(s + 1, false);
    queue<int> q;
    q.push(start);
    marked[start] = true;

    while (!q.empty())
    {
        int v = q.front();
        cout << v << " ";
        q.pop();

        for (int i = 0; i < graph[v].size(); i++)
        {
            int neigh = graph[v][i];
            if (marked[neigh] == false)
            {
                marked[neigh] = true;
                q.push(neigh);
            }
        }
    }
    cout << endl;
}

int main()
{
    cout << "Node and Edge : ";
    int n, e, m;
    cin >> n >> e;

    vector<int> graph[n + 1];
    for (int i = 1; i <= e; i++)
    {
        int a, b;
        cin >> a >> b;
        graph[a].push_back(b);
        graph[b].push_back(a);
    }

    cout << "Start node : ";
    cin >> m;

    BFS(graph, m, n);

    return 0;
}
```

Source Code (Task 2: Depth-First Search (DFS)):

```
#include<bits/stdc++.h>
using namespace std;
void DFS(vector<int>v[],int start,vector<bool>&visited)
{
    cout<<start<<" ";
    visited[start]=true;
    for(int i=0;i<v[start].size();i++)
    {
        int neigh=v[start][i];
        if(visited[neigh]==false)
        {
            DFS(v,neigh,visited);
        }
    }
}
int main()
{
    cout<<"Enter node and edge : ";
    int n,e,m;
    cin>>n>>e;
    vector<int>g[n+1];
    for(int i=0;i<e;i++)
    {
        int a,b;
        cin>>a>>b;
        g[a].push_back(b);
        g[b].push_back(a);
    }
    vector<bool>visited(n+1,false);
    cout<<"Enter start node : ";
    cin>>m;
    DFS(g,m,visited);
}
```

Execution Trace (Input & Output):

1. Breadth-First Search (BFS) Output:

```
"C:\Users\Hasan\Data Structu  X + v
Node and Edge : 5 4
1 2
1 3
2 4
3 5
Start node : 1
1 2 3 4 5

Process returned 0 (0x0)    execution time : 21.131 s
Press any key to continue.
|
```

2. Depth-First Search (DFS) Output:

```
Enter node and edge : 5 4
1 2
1 3
2 4
3 5
Enter start node : 1
1 2 4 3 5
Process returned 0 (0x0)    execution time : 10.628 s
Press any key to continue.
|
```